



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Assignment 1: Normalization

SECP2523 - DATABASE WBL

SEMESTER I, SESSION 2024/2025

SECTION 01

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Question 1

- a) FD1: medID, patientID, date $\rightarrow$ refillAllowed, PatientName, dosage, medName, shelfLife, doctorID, doctorName
FD2: medID $\rightarrow$ medName, shelfLife
FD3: patientID $\rightarrow$ PatientName
FD4: doctorID $\rightarrow$ doctorName
- b) 2NF Relations
- From FD1: R1(medID, medName, shelfLife)
Primary Key: medID
- From FD2: R2(patientID, PatientName)
Primary Key: patientID
- From FD3: R3(doctorID, doctorName)
Primary Key: doctorID
- FD4: R4(medID, patientID, date, refillAllowed, dosage, doctorID)
Primary Key: medID, patientID, date, doctorID
Foreign Key: medID references R1(medID)
Foreign Key: patientID references R2(patientID)
Foreign Key: doctorID references R3(doctorID)
- c) There is no transitive dependencies

Question 2

- a) FD1: $A, B \rightarrow E, F, G, H, I$
FD2: $C \rightarrow D$
FD3: $A \rightarrow E, F, G$

- b) 2NF Relations:
From FD2: R1(C, D)
Primary key: C
From FD3: R2(A, E, F, G)
Primary key: A
The rest: R3(A, B, C, H, I)
Primary key: A, B, C
Foreign key: A references R2(A)
Foreign key: C references R1(C)

- c) No transitive dependencies.

3NF Relations:
R1(C, D)
Primary key: C

R2(A, E, F, G)
Primary key: A

R3(A, B, C, H, I)
Primary key: A, B, C
Foreign key: A references R2(A)
Foreign key: C references R1(C)

Question 3

a) FD1: $M, N \rightarrow O, P, Q, R, S, T, U$

FD2: $N \rightarrow Q$

FD3: $O \rightarrow P$

FD4: $U \rightarrow T$

b) 2NF:

From FD2: R1 (N, Q)

Primary Key: N

From FD1: R (M, N, O, P, R, S, T, U)

Primary Key: M, N

Foreign Key: N references R1 (N)

c) 3NF:

R1 (N, Q)

Primary Key: N

R2 (O, P)

Primary Key: O

R3 (U, T)

Primary Key: U

R (M, N, O, R, S, U)

Primary Key: M, N

Foreign Key: N references R1 (N)

Foreign Key: O references R2 (O)

Foreign Key: U references R3 (U)

Question 4

- a) FD1: PropertyID $\rightarrow$ County, LotNo, Area, Price, TaxRate
FD2: County, LotNo $\rightarrow$ PropertyID, Area, Price, TaxRate
FD3: County $\rightarrow$ TaxRate

- b) LOTS(PropertyID, County, LotNo, Area, Price, TaxRate)
Primary key: PropertyID

LOTS is currently in 2NF because other non-primary key attributes are fully dependent on the primary key.

- c) LOTS_FOR_SALE(PropertyID, County, LotNo, Area, Price)
Primary key: PropertyID

LOTS_BASE(PropertyID, County, LotNo, Area, Price, TaxRate)
Primary key: PropertyID

CountyTaxRate(County, TaxRate)
Primary key: County